Chapter 3. Inference for Contingency Tables

Chi-squared tests, ordinal trends, exact tests

Contents

3.1 Confidence intervals for association parameters **3**
- 3.1.1 Interval estimation of odds ratios . . . 3
- 3.1.2 Aspirin and myocardial infarction example . . . 3
- 3.1.3 Interval estimation of difference of proportions . . . 3
- 3.1.4 Interval estimation of relative risk . . . 4
- 3.1.5 Deriving standard errors with the delta method* . . . 4
- 3.1.6 Delta method applied to sample logit* . . . 5
- 3.1.7 Delta method for log odds ratio* . . . 6
- 3.1.8 Score and profile likelihood confidence intervals* . . . 6

3.2 Testing independence in two-way contingency tables **7**
- 3.2.1 Pearson and likelihood-ratio chi-squared tests . . . 7
- 3.2.2 Education and religious fundamentalism example . . . 8

3.3 Following-up chi-squared tests **8**
- 3.3.1 Pearson and standardized residuals . . . 8
- 3.3.2 Education and religious fundamentalism revisited . . . 9
- 3.3.3 Partitioning chi-squared . . . 9
- 3.3.4 Origin of schizophrenia example . . . 10
- 3.3.5 Rules for partitioning . . . 11
- 3.3.6 Limitations of chi-squared tests . . . 11
- 3.3.7 Why consider independence? . . . 11

3.4 Two-way tables with ordered classifications **12**
- 3.4.1 Linear trend alternative to independence . . . 12
- 3.4.2 Job satisfaction example revisited . . . 13
- 3.4.3 Monotone trend alternatives to independence . . . 13
- 3.4.4 Extra power with ordinal tests . . . 13
- 3.4.5 Choice of scores . . . 13
- 3.4.6 Trend tests for $I \times 2$ and $2 \times J$ tables . . . 14
- 3.4.7 Nominal–ordinal tables . . . 14

3.5 Small-sample tests of independence **14**
- 3.5.1 Fisher's exact test for 2×2 tables . . . 14
- 3.5.2 Fisher's tea drinker . . . 15
- 3.5.3 Two-sided P-values for Fisher's exact test . . . 16
- 3.5.4 Discreteness and conservatism issues . . . 16
- 3.5.5 Small-sample unconditional test of independence* . . . 16
- 3.5.6 Conditional versus unconditional tests* . . . 17
- 3.5.7 Derivation of exact conditional distribution* . . . 17
- 3.5.8 Exact tests of independence for $I \times J$ tables* . . . 18

3.6 Small-sample confidence intervals for 2×2 tables* **18**
- 3.6.1 Small-sample inference for the odds ratio . . . 18
- 3.6.2 Tea tasting example . . . 20
- 3.6.3 Impact of discreteness on exact confidence intervals . . . 20
- 3.6.4 Small-sample inference for difference of proportions . . . 20

3.7 Extensions for multiway tables and nontabulated responses **20**
3.7.1 Categorical data need not be contingency tables 20

3.8 R lab **20**

3.1 Confidence intervals for association parameters

Motivation. Point estimates of the odds ratio, relative risk or difference of proportions mean little without a measure of precision. Large-sample standard errors come from one general tool, the delta method, and better intervals come from inverting score or likelihood-ratio tests.

3.1.1 Interval estimation of odds ratios

The sample odds ratio $\hat{\theta} = n_{11}n_{22}/(n_{12}n_{21})$ equals 0 or ∞ if some $n_{ij} = 0$, and is undefined if a whole row or column is zero. These events have positive probability, so $E(\hat{\theta})$ and $\mathrm{Var}(\log\hat{\theta})$ do not exist. The amended estimator

$$\tilde{\theta} = \frac{(n_{11}+0.5)(n_{22}+0.5)}{(n_{12}+0.5)(n_{21}+0.5)}$$

(and $\log\tilde{\theta}$) behaves well in terms of bias and mean squared error; $\hat{\theta}$ and $\tilde{\theta}$ have the same asymptotic normal distribution. Unless n is large the distribution of $\hat{\theta}$ is very skewed (for $\theta = 1$ it cannot be much below 1 but can be much above), whereas $\log\hat{\theta}$, with additive structure, approaches normality faster. Its estimated standard error is (Corollary 3.7)

$$\hat{\sigma}(\log\hat{\theta}) = \Big(\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}\Big)^{1/2}. \tag{3.1}$$

Definition (Woolf interval)

$\log\hat{\theta} \pm z_{\alpha/2}\,\hat{\sigma}(\log\hat{\theta})$ is a Wald interval for $\log\theta$; exponentiating its endpoints gives the Woolf interval for θ.

It works well, usually slightly conservatively. When $\hat{\theta} = 0$ (or ∞) take 0 as the lower (or ∞ as the upper) limit and compute the other bound after an adjustment such as replacing n_{ij} by $n_{ij} + 0.5$; a less ad hoc approach inverts score or LR tests (Section 3.1.8).

3.1.2 Aspirin and myocardial infarction example

Example 3.1 (Swedish aspirin study: odds ratio)

1360 stroke patients were randomized to low-dose aspirin or placebo and followed about three years for death from myocardial infarction (MI).

	MI death		
	Yes	No	Total
Placebo	28	656	684
Aspirin	18	658	676

$\hat{\theta} = \frac{28\times658}{656\times18} = 1.56$ (close to $\tilde{\theta} = 1.55$), $\log\hat{\theta} = 0.445$, and by (3.1) $\hat{\sigma}(\log\hat{\theta}) = (1/28 + 1/656 + 1/18 + 1/658)^{1/2} = 0.307$. The 95% interval for $\log\theta$ is $0.445 \pm 1.96(0.307) = (-0.157,\ 1.047)$, so for θ it is $(e^{-0.157}, e^{1.047}) = (0.85,\ 2.85)$. Since it contains 1, equal odds of MI death for aspirin and placebo are plausible; a small benefit would need a large sample to detect.

3.1.3 Interval estimation of difference of proportions

Treat the rows as independent binomials: $y_i \sim \mathrm{bin}(n_i, \pi_i)$, $\hat{\pi}_i = y_i/n_i$.

Proposition 3.2 (Difference of proportions)

$E(\hat{\pi}_1 - \hat{\pi}_2) = \pi_1 - \pi_2$ and

$$\sigma(\hat{\pi}_1 - \hat{\pi}_2) = \left[\frac{\pi_1(1-\pi_1)}{n_1} + \frac{\pi_2(1-\pi_2)}{n_2}\right]^{1/2}. \tag{3.2}$$

As $n_1, n_2 \to \infty$, $(\hat{\pi}_1 - \hat{\pi}_2 - (\pi_1 - \pi_2))/\hat{\sigma} \xrightarrow{d} N(0,1)$, where $\hat{\sigma}$ replaces π_i by $\hat{\pi}_i$; so $(\hat{\pi}_1 - \hat{\pi}_2) \pm z_{\alpha/2}\hat{\sigma}$ is a Wald interval.

Proof.

Step 1. Linearity and independence give the mean and variance, using $\text{Var}(\hat{\pi}_i) = \pi_i(1-\pi_i)/n_i$ (Proposition 1.8).

Step 2. [Supplement] Each $(\hat{\pi}_i - \pi_i)/\sigma(\hat{\pi}_i) \xrightarrow{d} N(0,1)$ by the CLT, independently, so the standardized difference is asymptotically $N(0,1)$; $\hat{\sigma}/\sigma \xrightarrow{p} 1$ by the WLLN, and Slutsky's theorem completes the proof. ■

Like the single-proportion Wald interval (1.11), its coverage is usually below nominal, especially when π_1, π_2 are near 0 or 1 (adding one success and one failure to each sample helps greatly).

3.1.4 Interval estimation of relative risk

The sample relative risk $r = \hat{\pi}_1/\hat{\pi}_2$ also converges to normality faster on the log scale, with asymptotic standard error

$$\sigma(\log r) = \left[\frac{1-\pi_1}{\pi_1 n_1} + \frac{1-\pi_2}{\pi_2 n_2}\right]^{1/2} \tag{3.3}$$

(Corollary 3.5(b)). The Wald interval exponentiates $\log r \pm z_{\alpha/2}\hat{\sigma}(\log r)$; it works well but can be somewhat conservative.

Example 3.3 (Swedish aspirin study: relative risk and difference)

Sample MI death proportions are $28/684 = 0.0409$ (placebo) and $18/676 = 0.0266$ (aspirin).
(a) $r = 1.54$; $\log 1.54 \pm 1.96(0.297)$ gives (0.86, 2.75) for the relative risk.
(b) Difference: $0.014 \pm 1.96(0.0098)$, i.e. $(-0.005, 0.033)$.
Substantial benefit is plausible, but so are no effect and a slight harm; the larger Physicians' Health Study (Example 2.7) does show a benefit.

3.1.5 Deriving standard errors with the delta method*

Theorem 3.4 (Delta method)

Suppose $\sqrt{n}(T_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ and g is differentiable at θ. Then

$$\sqrt{n}\,[g(T_n) - g(\theta)] \xrightarrow{d} N(0,\, [g'(\theta)]^2\sigma^2).$$

If moreover g' is continuous at θ, $g'(\theta) \neq 0$, and $\hat{\sigma} \xrightarrow{p} \sigma > 0$, then $\sqrt{n}[g(T_n) - g(\theta)]/(|g'(T_n)|\hat{\sigma}) \xrightarrow{d} N(0,1)$.

Proof. [Supplement]

Step 1. Define $h(t) = \frac{g(t)-g(\theta)}{t-\theta} - g'(\theta)$ for $t \neq \theta$ and $h(\theta) = 0$; differentiability means h is continuous at θ.

Step 2. $T_n - \theta = n^{-1/2} \cdot \sqrt{n}(T_n - \theta) = o_p(1)$, so $T_n \xrightarrow{p} \theta$ and $h(T_n) \xrightarrow{p} 0$ (continuous mapping).

Step 3. Then $\sqrt{n}[g(T_n) - g(\theta)] = [g'(\theta) + h(T_n)]\sqrt{n}(T_n - \theta)$. The bracket $\xrightarrow{p} g'(\theta)$, so by Slutsky the product $\xrightarrow{d} g'(\theta)N(0, \sigma^2) = N(0, g'(\theta)^2\sigma^2)$ (recall $cY \sim N(0, c^2\sigma^2)$ when $Y \sim N(0, \sigma^2)$).

Step 4. $|g'(T_n)|\hat{\sigma} \xrightarrow{p} |g'(\theta)|\sigma > 0$, and a further application of Slutsky gives the standardized version. ■

Locally g is approximately linear with slope $g'(\theta)$, so $g(T_n)$ is approximately normal and its spread around $g(\theta)$ is $|g'(\theta)|$ times the spread of T_n around θ (Figure 1). A large-sample 95% Wald interval for $g(\theta)$ is $g(T_n) \pm 1.96|g'(T_n)|\hat{\sigma}/\sqrt{n}$.

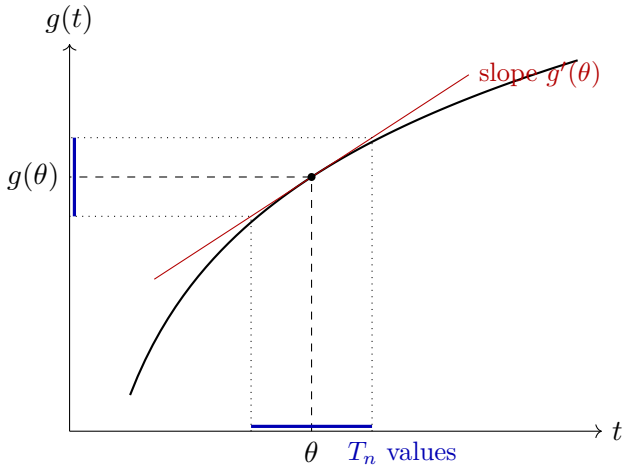


Figure 1: The delta method: near θ, g is almost linear, so a band of T_n values maps to a band of $g(T_n)$ values about $|g'(\theta)|$ times as wide.

3.1.6 Delta method applied to sample logit*

Corollary 3.5 (Sample logit and log proportion)

Let $\hat{\pi} = y/n$ with $y \sim \text{bin}(n, \pi)$, $0 < \pi < 1$.
(a) The sample logit satisfies

$$\sqrt{n}\Big[\log\frac{\hat{\pi}}{1-\hat{\pi}} - \log\frac{\pi}{1-\pi}\Big] \xrightarrow{d} N\Big(0, \frac{1}{\pi(1-\pi)}\Big).$$

(b) $\sqrt{n}(\log\hat{\pi} - \log\pi) \xrightarrow{d} N(0, (1-\pi)/\pi)$; hence for independent samples the log relative risk has asymptotic variance (3.3).

Proof.

Step (a). $\sqrt{n}(\hat{\pi} - \pi) \xrightarrow{d} N(0, \pi(1-\pi))$ by the CLT. With $g(\pi) = \log[\pi/(1-\pi)]$, $g'(\pi) = 1/\pi + 1/(1-\pi) = 1/[\pi(1-\pi)]$; Theorem 3.4 gives variance $\pi(1-\pi) \cdot [\pi(1-\pi)]^{-2}$.

Step (b). With $g = \log$, $g'(\pi) = 1/\pi$ and the variance is $\pi(1-\pi)/\pi^2 = (1-\pi)/\pi$. Dividing by n_i and adding over the two independent samples gives (3.3) for $\log r = \log\hat{\pi}_1 - \log\hat{\pi}_2$. ■

The *asymptotic* variance $[n\pi(1-\pi)]^{-1}$ is the variance of the approximating normal distribution, not an approximation to the true variance—which does not exist, since $\hat{\pi} = 0$ or 1 (logit $\mp\infty$) with positive probability. That probability vanishes rapidly, so the asymptotic variance is the more useful quantity. For the same reason the bootstrap, which mimics the true sampling variance, is unhelpful for SEs of many discrete measures.

3.1.7 Delta method for log odds ratio*

Theorem 3.6 (Multivariate delta method for multinomial proportions)

Let $(n_1, \ldots, n_c)$ be multinomial$(n, \boldsymbol{\pi})$, $\hat{\boldsymbol{\pi}} = \mathbf{n}/n$, and let g be differentiable at $\boldsymbol{\pi}$ with $\phi_i = \partial g/\partial \pi_i$. Define

$$\sigma^2 = \sum_i \pi_i \phi_i^2 - \Big(\sum_i \pi_i \phi_i\Big)^2. \tag{3.4}$$

If $\sigma > 0$ then $\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})]/\sigma \xrightarrow{d} N(0,1)$, and the same holds with σ replaced by its ML estimate $\hat{\sigma}$ (substituting $\hat{\pi}_i$ for π_i in (3.4)).

Proof. [Supplement]

Step 1. By the multivariate CLT, $\sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) \xrightarrow{d} \mathbf{z} \sim N_c(\mathbf{0}, \Sigma)$ with $\Sigma = \operatorname{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top$ (Proposition 1.2).

Step 2. As in Theorem 3.4, differentiability gives $g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi}) = \boldsymbol{\phi}^\top(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) + o_p(\|\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}\|)$, and $\|\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}\| = O_p(n^{-1/2})$. Hence $\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})] = \boldsymbol{\phi}^\top \sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) + o_p(1) \xrightarrow{d} \boldsymbol{\phi}^\top \mathbf{z}$.

Step 3. $\boldsymbol{\phi}^\top \mathbf{z} \sim N(0, \boldsymbol{\phi}^\top \Sigma \boldsymbol{\phi})$ and $\boldsymbol{\phi}^\top \Sigma \boldsymbol{\phi} = \sum_i \pi_i \phi_i^2 - (\sum_i \pi_i \phi_i)^2 = \sigma^2$.

Step 4. By the WLLN $\hat{\boldsymbol{\pi}} \xrightarrow{p} \boldsymbol{\pi}$; if the ϕ_i are continuous, $\hat{\sigma} \xrightarrow{p} \sigma$ and $\sigma/\hat{\sigma} \xrightarrow{p} 1$. Writing

$$\frac{\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})]}{\hat{\sigma}} = \frac{\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})]}{\sigma} \cdot \frac{\sigma}{\hat{\sigma}},$$

Slutsky's theorem gives the limit $N(0,1)$ (convergence is slower with $\hat{\sigma}$). ■

Corollary 3.7 (Standard error of the log odds ratio)

For a multinomial 2×2 table, $\log \hat{\theta}$ has asymptotic standard error $\big(\sum_i \sum_j 1/(n\pi_{ij})\big)^{1/2}$, estimated by (3.1).

Proof.

Step 1. $g(\boldsymbol{\pi}) = \log \pi_{11} + \log \pi_{22} - \log \pi_{12} - \log \pi_{21}$ has $\phi_{11} = 1/\pi_{11}$, $\phi_{12} = -1/\pi_{12}$, $\phi_{21} = -1/\pi_{21}$, $\phi_{22} = 1/\pi_{22}$.

Step 2. $\sum\sum \pi_{ij}\phi_{ij} = 1 - 1 - 1 + 1 = 0$ and $\sum\sum \pi_{ij}\phi_{ij}^2 = \sum\sum 1/\pi_{ij}$. By Theorem 3.6, $\sigma(\log\hat{\theta}) = \sigma/\sqrt{n} = \big(\sum\sum 1/(n\pi_{ij})\big)^{1/2}$, and substituting $\hat{\pi}_{ij} = n_{ij}/n$ gives $\big(\sum\sum 1/n_{ij}\big)^{1/2}$. ■

The same result holds for independent binomial rows (Corollary 3.5(a): $\log\hat{\theta}$ is a difference of two independent sample logits, with variance $\frac{1}{n_1\pi_1(1-\pi_1)} + \frac{1}{n_2\pi_2(1-\pi_2)}$, estimated by (3.1)). A Wald interval for θ itself, $\hat{\theta} \pm z_{\alpha/2}\hat{\sigma}(\hat{\theta})$, is not recommended: $\hat{\theta}$ approaches normality slowly, the interval can include negative values, and it is not equivalent to the interval for $1/\theta$.

3.1.8 Score and profile likelihood confidence intervals*

Wald intervals can perform poorly for small to moderate n; inverting score or LR tests usually performs better.

(a) *Score interval for* $\pi_1 - \pi_2$ (Mee; Miettinen and Nurminen): for $H_0\colon \pi_1 - \pi_2 = \Delta$,

$$z(\Delta) = \frac{\hat{\pi}_1 - \hat{\pi}_2 - \Delta}{\big[\hat{\pi}_1(\Delta)(1-\hat{\pi}_1(\Delta))/n_1 + \hat{\pi}_2(\Delta)(1-\hat{\pi}_2(\Delta))/n_2\big]^{1/2}},$$

where $\hat{\pi}_i(\Delta)$ maximize the product binomial likelihood subject to $\pi_1 - \pi_2 = \Delta$ (found numerically). The interval is $\{\Delta : |z(\Delta)| < z_{\alpha/2}\}$. Analogous score intervals exist for the relative risk and the odds ratio (Cornfield). Continuity corrections make them too conservative.

(b) *Profile likelihood interval for θ*: the multinomial likelihood can be reparametrized as $(\theta, \pi_{1+}, \pi_{+1})$ (Section 2.4.1). For each θ_0 let $\hat{\pi}_{1+}(\theta_0), \hat{\pi}_{+1}(\theta_0)$ maximize the likelihood with $\theta = \theta_0$; the *profile log likelihood* is $L(\theta_0, \hat{\pi}_{1+}(\theta_0), \hat{\pi}_{+1}(\theta_0))$. The interval is

$$\{\theta_0 : -2[L(\theta_0, \hat{\pi}_{1+}(\theta_0), \hat{\pi}_{+1}(\theta_0)) - L(\hat{\theta}, \hat{\pi}_{1+}, \hat{\pi}_{+1})] < \chi^2_1(\alpha)\},$$

the θ_0 not rejected by LR tests of size α; the unrestricted maximum is at n_{1+}/n, n_{+1}/n.

Unlike Wald intervals, score and LR intervals are not damaged when $\hat{\theta}$ or r is 0 or ∞. A conditional likelihood (Section 6.7.1) instead eliminates nuisance parameters by conditioning on their sufficient statistics.

3.2 Testing independence in two-way contingency tables

Motivation. The null hypothesis of no association, $\pi_{ij} = \pi_{i+}\pi_{+j}$, is the starting point of contingency-table analysis. Pearson's X^2 and the LR statistic G^2 test it, and getting their degrees of freedom right was the occasion for Fisher's introduction of the concept.

For multinomial sampling H_0: $\pi_{ij} = \pi_{i+}\pi_{+j}$ for all i, j; for independent multinomial rows this is homogeneity of the conditional distributions. The same tests apply in both cases.

3.2.1 Pearson and likelihood-ratio chi-squared tests

Proposition 3.8 (ML estimates under independence)

Under H_0, the ML estimates are $\hat{\pi}_{i+} = n_{i+}/n$, $\hat{\pi}_{+j} = n_{+j}/n$, so the estimated expected frequencies are

$$\hat{\mu}_{ij} = n\hat{\pi}_{i+}\hat{\pi}_{+j} = \frac{n_{i+}n_{+j}}{n}.$$

Proof. Under H_0 the multinomial kernel is $\prod_i \prod_j (\pi_{i+}\pi_{+j})^{n_{ij}} = \prod_i \pi_{i+}^{n_{i+}} \cdot \prod_j \pi_{+j}^{n_{+j}}$, a product of two multinomial kernels in separate parameters. Each is maximized by the sample proportions (Proposition 1.16). ■

$$X^2 = \sum_i \sum_j \frac{(n_{ij} - \hat{\mu}_{ij})^2}{\hat{\mu}_{ij}}, \qquad G^2 = 2\sum_i \sum_j n_{ij} \log \frac{n_{ij}}{\hat{\mu}_{ij}}. \tag{3.5}$$

G^2 is $-2\log\Lambda$ with $\Lambda = \prod\prod (n_{i+}n_{+j})^{n_{ij}} / (n^n \prod\prod n_{ij}^{n_{ij}})$, the ratio of the kernel at $\hat{\pi}_{ij} = n_{i+}n_{+j}/n^2$ to that at $\hat{\pi}_{ij} = n_{ij}/n$.

Theorem 3.9 (Null distribution of X^2 and G^2 for independence)

Under H_0 with all $\pi_{i+}, \pi_{+j} > 0$, X^2 and G^2 both converge in distribution to $\chi^2_{(I-1)(J-1)}$, and $X^2 - G^2 \xrightarrow{p} 0$.

Proof. [Supplement] Apply Theorem 1.20 to the $c = IJ$ cells with $p = (I-1) + (J-1)$ parameters

$$\boldsymbol{\theta} = (\pi_{1+}, \ldots, \pi_{I-1,+}, \pi_{+1}, \ldots, \pi_{+,J-1}).$$

Step 1 (rank). With $a_i = \pi_{i+}$, $b_j = \pi_{+j}$, a tangent direction is $d\pi_{ij} = b_j\,da_i + a_i\,db_j$ with $\sum_i da_i = \sum_j db_j = 0$. If $d\pi_{ij} = 0$ for all i, j, summing over j gives $da_i = 0$, and then $a_i\,db_j = 0$ gives $db_j = 0$. So the Jacobian has full rank p.

Step 2 (df). Hence $X^2 \xrightarrow{d} \chi^2_{IJ-1-p}$ with

$$IJ - 1 - (I-1) - (J-1) = IJ - I - J + 1 = (I-1)(J-1),$$

and $G^2 - X^2 \xrightarrow{p} 0$ by the same theorem. ■

Pearson originally claimed df $= IJ - 1$; Fisher (1922) corrected him, introducing the notion of degrees of freedom. Equivalently, the general parameter space has dimension $IJ - 1$ and the null space $(I-1) + (J-1)$; their difference is the df. The limits also hold for Poisson and product multinomial sampling. Convergence is quicker for X^2 than G^2; the approximation for G^2 is often poor when $n/(IJ) < 5$, while X^2 can be adequate with some $\hat{\mu}_{ij}$ near 1 if most exceed 5. Otherwise use small-sample methods (Section 3.5).

3.2.2 Education and religious fundamentalism example

Example 3.10 (Education and religious beliefs)

1996 General Social Survey. Each cell shows the count, the estimate $\hat{\mu}_{ij}$ under independence, and the standardized Pearson residual (3.6).

Highest degree	Fundamentalist		Moderate		Liberal		Total
< High school	178	(137.8) 4.5	138	(161.5) −2.6	108	(124.7) −1.9	424
HS / junior college	570	(539.5) 2.6	648	(632.1) 1.3	442	(488.4) −4.0	1660
Bachelor / graduate	138	(208.7) −6.8	252	(244.5) 0.7	252	(188.9) 6.3	642
Total	886		1038		802		2726

E.g. $\hat{\mu}_{11} = 424 \times 886/2726 = 137.8$. Here $X^2 = 69.2$ and $G^2 = 69.8$ with df $= (3-1)(3-1) = 4$, $P < 0.0001$: extremely strong evidence of association.

3.3 Following-up chi-squared tests

Motivation. A small P-value says that an association exists, not what it looks like or how strong it is. Residuals, partitions of G^2 and estimated odds ratios reveal its nature.

3.3.1 Pearson and standardized residuals

Raw differences $n_{ij} - \hat{\mu}_{ij}$ tend to be larger where $\hat{\mu}_{ij}$ is larger (under Poisson sampling the standard deviation of n_{ij} is $\sqrt{\mu_{ij}}$).

Definition (Pearson and standardized residuals)

The *Pearson residual* and *standardized Pearson residual* for independence are

$$e_{ij} = \frac{n_{ij} - \hat{\mu}_{ij}}{\hat{\mu}_{ij}^{1/2}}, \qquad r_{ij} = \frac{n_{ij} - \hat{\mu}_{ij}}{[\hat{\mu}_{ij}(1 - p_{i+})(1 - p_{+j})]^{1/2}}, \tag{3.6}$$

with $p_{i+} = n_{i+}/n$, $p_{+j} = n_{+j}/n$. Note $\sum_i \sum_j e_{ij}^2 = X^2$.

Theorem 3.11 (Asymptotic variance of Pearson residuals)

Under independence, $e_{ij} \xrightarrow{d} N(0,(1-\pi_{i+})(1-\pi_{+j}))$; hence $r_{ij} \xrightarrow{d} N(0,1)$. The average asymptotic variance of the e_{ij} is $(I-1)(J-1)/(IJ) < 1$.

Proof. [Supplement] Use the notation of the proof of Theorem 1.20, indexing cells by (i,j). Let $a_i = \pi_{i+}$, $b_j = \pi_{+j}$, and the unit vectors $\mathbf{r} = (\sqrt{a_i})_i \in \mathbb{R}^I$, $\mathbf{s}_c = (\sqrt{b_j})_j \in \mathbb{R}^J$; then $\mathbf{s} = (\sqrt{\pi_{ij}}) = \mathbf{r} \otimes \mathbf{s}_c$.

Step 1. By that proof the vector of Pearson residuals is $(I-H)\mathbf{z}_n + o_p(1)$ with limiting covariance $P = I - H - \mathbf{s}\mathbf{s}^\top$, where H is the projection onto the column space of $A = D^{-1/2}\partial\boldsymbol{\pi}/\partial\boldsymbol{\theta}^\top$.

Step 2. A tangent direction $d\pi_{ij} = b_j\,da_i + a_i\,db_j$ (with $\sum da_i = \sum db_j = 0$) gives $D^{-1/2}d\boldsymbol{\pi}$ with entries $\sqrt{b_j}\,x_i + \sqrt{a_i}\,y_j$, where $x_i = da_i/\sqrt{a_i}$, $y_j = db_j/\sqrt{b_j}$; the constraints say $\mathbf{x} \perp \mathbf{r}$, $\mathbf{y} \perp \mathbf{s}_c$. So the column space of A is $(\mathbf{r}^\perp \otimes \mathbf{s}_c) \oplus (\mathbf{r} \otimes \mathbf{s}_c^\perp)$, two orthogonal subspaces, and

$$H = (I_I - \mathbf{r}\mathbf{r}^\top) \otimes \mathbf{s}_c\mathbf{s}_c^\top + \mathbf{r}\mathbf{r}^\top \otimes (I_J - \mathbf{s}_c\mathbf{s}_c^\top).$$

Step 3. The (ij,ij) diagonal entry of H is $(1-a_i)b_j + a_i(1-b_j)$ and that of $\mathbf{s}\mathbf{s}^\top$ is a_ib_j. Hence the asymptotic variance of e_{ij} is

$$1 - a_ib_j - (1-a_i)b_j - a_i(1-b_j) = 1 - a_i - b_j + a_ib_j = (1-a_i)(1-b_j).$$

Step 4. Since $p_{i+} \xrightarrow{p} a_i$ and $p_{+j} \xrightarrow{p} b_j$, Slutsky gives $r_{ij} = e_{ij}/[(1-p_{i+})(1-p_{+j})]^{1/2} \xrightarrow{d} N(0,1)$. Finally $\frac{1}{IJ}\sum_i\sum_j(1-a_i)(1-b_j) = \frac{(I-1)(J-1)}{IJ}$. ∎

So comparing e_{ij} with $N(0,1)$ is conservative, while $|r_{ij}|$ above about 2 or 3 indicates lack of fit of H_0 in that cell (larger thresholds when df is large, since some large values then occur by chance).

3.3.2 Education and religious fundamentalism revisited

Example 3.12 (Residuals for education and religion)

For the first cell $p_{1+} = 424/2726 = 0.156$, $p_{+1} = 886/2726 = 0.325$ and

$$r_{11} = \frac{178 - 137.8}{[137.8(1-0.156)(1-0.325)]^{1/2}} = 4.5.$$

Large positive residuals occur for (less than high school, fundamentalist) and (bachelor or graduate, liberal), large negative ones for (bachelor or graduate, fundamentalist) and (less than high school, liberal). The corner 2×2 table has odds ratio $\frac{178\times 252}{108\times 138} = 3.0$: for the most educated, the odds of liberal rather than fundamentalist were 3 times those for the least educated.

3.3.3 Partitioning chi-squared

If $Z_1,\ldots,Z_\nu$ are i.i.d. $N(0,1)$ then $Z_1^2 + \cdots + Z_\nu^2 \sim \chi^2_\nu$; so a χ^2_ν variable splits into independent components, and conversely independent $\chi^2_{\nu_1}$ and $\chi^2_{\nu_2}$ variables sum to $\chi^2_{\nu_1+\nu_2}$. A partition of a test statistic can show which categories drive an association.

Theorem 3.13 (Exact partition of G^2 by successive columns)

For an $I \times J$ table, for $j = 1, \ldots, J-1$ let G_j^2 be the G^2 statistic for independence in the $I \times 2$ table whose first column merges columns $1, \ldots, j$ and whose second column is column $j+1$. Then

$$G^2 = \sum_{j=1}^{J-1} G_j^2,$$

and each G_j^2 has df $= I - 1$, so the df also add up to $(I-1)(J-1)$.

Proof. [Supplement]

Step 1. For any table with counts m_{ij}, row totals m_{i+}, column totals m_{+j} and total m, (3.5) gives

$$G^2 = 2\Big[\sum_i \sum_j m_{ij} \log m_{ij} - \sum_i m_{i+} \log m_{i+} - \sum_j m_{+j} \log m_{+j} + m \log m\Big],$$

since $\log(m_{ij}/\hat{\mu}_{ij}) = \log m_{ij} + \log m - \log m_{i+} - \log m_{+j}$. Write $\Phi(x) = x \log x$.

Step 2. Let $S_{ij} = \sum_{l \le j} n_{il}$ (so $S_{i1} = n_{i1}$, $S_{iJ} = n_{i+}$) and $S_{+j} = \sum_i S_{ij}$. The jth subtable has cells S_{ij}, $n_{i,j+1}$, row totals $S_{i,j+1}$, column totals S_{+j}, $n_{+,j+1}$ and total $S_{+,j+1}$. Hence $G_j^2/2$ equals

$$\sum_i [\Phi(S_{ij}) + \Phi(n_{i,j+1}) - \Phi(S_{i,j+1})] - \Phi(S_{+j}) - \Phi(n_{+,j+1}) + \Phi(S_{+,j+1}).$$

Step 3. Sum over $j = 1, \ldots, J-1$. The terms $\Phi(S_{ij}) - \Phi(S_{i,j+1})$ telescope to $\Phi(n_{i1}) - \Phi(n_{i+})$, and adding $\sum_{j \ge 2} \Phi(n_{ij})$ gives $\sum_j \Phi(n_{ij}) - \Phi(n_{i+})$. Likewise $\sum_j [\Phi(S_{+,j+1}) - \Phi(S_{+j})] - \sum_{j \ge 2} \Phi(n_{+j}) = \Phi(n) - \sum_j \Phi(n_{+j})$.

Step 4. The total is exactly $\sum\sum \Phi(n_{ij}) - \sum_i \Phi(n_{i+}) - \sum_j \Phi(n_{+j}) + \Phi(n) = G^2/2$ by Step 1. ■

Applying the theorem again to each $I \times 2$ component, with rows merged successively, gives a finer partition into $(I-1)(J-1)$ components with df $= 1$ (Lancaster), based on the 2×2 tables

$$\begin{matrix} \sum_{a<i} \sum_{b<j} n_{ab} & \sum_{a<i} n_{aj} \\ \sum_{b<j} n_{ib} & n_{ij} \end{matrix} \qquad i = 2, \ldots, I, \ j = 2, \ldots, J. \tag{3.7}$$

Comparing each column with the last one instead does *not* give components that are independent or sum to G^2 (the corresponding log-odds-ratio contrasts are not orthogonal). In Section 9.2.4 the partition is justified as a sequence of nested models.

3.3.4 Origin of schizophrenia example

Example 3.14 (School of psychiatric thought and origin of schizophrenia)

School	Biogenic	Environmental	Combination
Eclectic	90	12	78
Medical	13	1	6
Psychoanalytic	19	13	50

$G^2 = 23.04$, df $= 4$. Partitioning by (3.7):

Rows compared	Columns compared	G^2 (df = 1)
Ecl vs. Med	Bio vs. Env	0.29
Ecl vs. Med	(Bio+Env) vs. Com	1.36
(Ecl+Med) vs. Psy	Bio vs. Env	12.95
(Ecl+Med) vs. Psy	(Bio+Env) vs. Com	8.43
Sum		23.04

The eclectic and medical schools hardly differ. The psychoanalytic school more often ascribes a combined origin and, among biogenic/environmental choices, more often the environmental one.

3.3.5 Rules for partitioning

Necessary conditions for subtables to give independent components (Goodman; Lancaster):

(a) the subtable df sum to the df of the full table;

(b) each cell count of the full table is a cell count in exactly one subtable;

(c) each marginal total of the full table is a marginal total of exactly one subtable.

If the df sum correctly but the G^2 values do not, the components are not independent. Exact algebraic partitions hold for G^2 but not for X^2; the subtable X^2 values are still valid tests (asymptotically equivalent to G^2 under the null), and are safer for small counts.

3.3.6 Limitations of chi-squared tests

Chi-squared tests only measure evidence of association; one should also study residuals, partitions and estimated odds ratios. They need large samples, and since $\hat{\mu}_{ij}$ depend only on the margins, X^2 and G^2 are invariant to reordering rows or columns: they treat both variables as nominal. With ordinal variables, tests using the ordering are more appropriate (Section 3.4).

3.3.7 Why consider independence?

Independence rarely holds exactly, and with large n small P-values are expected. Yet if independence approximately holds, the model-based estimates $\hat{\pi}_{ij} = p_{i+}p_{+j}$ can beat the sample proportions p_{ij} in mean squared error: they may be biased but estimate fewer parameters, so have smaller variance.

Proposition 3.15 (MSE of the two estimators)

For multinomial sampling,

$$\sum_i \sum_j E(p_{ij} - \pi_{ij})^2 = \frac{1 - \sum_i \sum_j \pi_{ij}^2}{n}, \qquad \sum_i \sum_j E(p_{i+}p_{+j} - \pi_{ij})^2 \to \sum_i \sum_j (\pi_{i+}\pi_{+j} - \pi_{ij})^2.$$

Under independence, $n \sum \sum E(p_{i+}p_{+j} - \pi_{ij})^2 \to \sum_i \sum_j [\pi_{+j}^2 \pi_{i+}(1 - \pi_{i+}) + \pi_{i+}^2 \pi_{+j}(1 - \pi_{+j})]$.

Proof. [Supplement]

Step 1. p_{ij} is unbiased with $\mathrm{Var}(p_{ij}) = \pi_{ij}(1 - \pi_{ij})/n$; summing gives $(1 - \sum \sum \pi_{ij}^2)/n$.

Step 2. Write $a = p_{i+} - \pi_{i+}$, $b = p_{+j} - \pi_{+j}$; then

$$p_{i+}p_{+j} - \pi_{ij} = (\pi_{i+}\pi_{+j} - \pi_{ij}) + \pi_{+j}a + \pi_{i+}b + ab.$$

Since $E(a^2), E(b^2) = O(n^{-1})$ and $E(a^2b^2) \le [E(a^4)E(b^4)]^{1/2} = O(n^{-2})$ (fourth central moments of a binomial proportion are $O(n^{-2})$), all terms but the first vanish in mean square, giving the second limit.

Step 3. Under independence the first term is 0, and $nE(\pi_{+j}a+\pi_{i+}b)^2 = \pi_{+j}^2\pi_{i+}(1-\pi_{i+})+\pi_{i+}^2\pi_{+j}(1-\pi_{+j}) + 2\pi_{i+}\pi_{+j}\, n\,\mathrm{Cov}(a,b)$. Here $n\,\mathrm{Cov}(p_{i+},p_{+j}) = \pi_{ij} - \pi_{i+}\pi_{+j} = 0$ (indicator covariance), and the remaining terms are $o(n^{-1})$ by Cauchy–Schwarz. ■

Example 3.16 (A 3×3 illustration)

Let $\pi_{i+} = \pi_{+j} = 1/3$ and $\pi_{ij} = \frac{1}{9}[1 + \delta(i-2)(j-2)]$, $-1 < \delta < 1$: the corner cells are $(1\pm\delta)/9$, the others $1/9$; $\delta = 0$ is independence.
(a) At $\delta = 0$: $\sum\sum \pi_{ij}^2 = 1/9$, so the sample proportions have total MSE $8/(9n)$, while the independence estimator has total MSE $\approx 9\cdot 2\cdot \frac{1}{9}\cdot\frac{1}{3}\cdot\frac{2}{3}/n = 4/(9n)$: about half.
(b) For $\delta \neq 0$ the independence estimator is inconsistent: its total MSE tends to $4(\delta/9)^2 = 4\delta^2/81$, while that of p_{ij} tends to 0. Still, near independence it wins for moderately large n (e.g. $n = 500$ when $\delta = 0.1$).

3.4 Two-way tables with ordered classifications

Motivation. X^2 and G^2 ignore category order. When both variables are ordinal, a single-parameter description of the trend (a correlation or gamma) yields tests with one degree of freedom and much greater power against monotone alternatives.

3.4.1 Linear trend alternative to independence

Assign row scores $u_1 \le \cdots \le u_I$ and column scores $v_1 \le \cdots \le v_J$ (treating the scales as interval). The covariation $\sum_i\sum_j u_i v_j n_{ij}$, standardized, gives the sample correlation r between the scores over the n subjects:

$$r = \frac{\sum\sum u_i v_j n_{ij} - \left(\sum_i u_i n_{i+}\right)\left(\sum_j v_j n_{+j}\right)/n}{\left[\sum_i u_i^2 n_{i+} - \frac{(\sum_i u_i n_{i+})^2}{n}\right]^{1/2}\left[\sum_j v_j^2 n_{+j} - \frac{(\sum_j v_j n_{+j})^2}{n}\right]^{1/2}}.$$

Definition (Linear trend statistic)

$M^2 = (n-1)r^2$ (Mantel, 1963). Large values contradict independence; the P-value is the right-tail χ_1^2 probability. For one-sided alternatives use $M = \sqrt{n-1}\,r$ with the normal tail.

Theorem 3.17 (Null distribution of M^2)

Under independence of X and Y (multinomial sampling, each score variable nondegenerate), $\sqrt{n}\,r \xrightarrow{d} N(0,1)$; hence $M \xrightarrow{d} N(0,1)$ and $M^2 \xrightarrow{d} \chi_1^2$.

Proof. [Supplement] Let (U_k, V_k), $k = 1,\ldots,n$, be the score pairs of the n subjects: i.i.d., with U independent of V, means μ,ν and variances $\sigma^2,\tau^2 > 0$.

Step 1. The sample covariance is $S_{UV} = \frac{1}{n}\sum_k (U_k-\mu)(V_k-\nu) - (\bar U-\mu)(\bar V-\nu)$. The summands $W_k = (U_k-\mu)(V_k-\nu)$ are i.i.d. with $E(W_k) = E(U-\mu)E(V-\nu) = 0$ and $\mathrm{Var}(W_k) = E(U-\mu)^2E(V-\nu)^2 = \sigma^2\tau^2$, by independence. So $\sqrt{n}\cdot\frac{1}{n}\sum_k W_k \xrightarrow{d} N(0,\sigma^2\tau^2)$ (CLT).

Step 2. $\sqrt{n}(\bar{U}-\mu)(\bar{V}-\nu) = \sqrt{n}(\bar{U}-\mu)\cdot(\bar{V}-\nu) = O_p(1)o_p(1) \xrightarrow{p} 0$. Hence $\sqrt{n}S_{UV} \xrightarrow{d} N(0, \sigma^2\tau^2)$.

Step 3. $S_{UU} \xrightarrow{p} \sigma^2$ and $S_{VV} \xrightarrow{p} \tau^2$ (WLLN). By Slutsky, $\sqrt{n}\, r = \sqrt{n}S_{UV}/(S_{UU}S_{VV})^{1/2} \xrightarrow{d} N(0,1)$.

Step 4. $\sqrt{(n-1)/n} \to 1$, so $M = \sqrt{n-1}\, r \xrightarrow{d} N(0,1)$ and $M^2 \xrightarrow{d} \chi^2_1$. ■

A small P-value does not mean the association is linear, only that looking for a linear component helped power. The test is symmetric in X and Y.

3.4.2 Job satisfaction example revisited

Example 3.18 (Linear trend test for income and job satisfaction)

For Example 2.17 ($n = 96$), $X^2 = 6.0$ and $G^2 = 6.8$ with df $= 9$ ($P = 0.74$, 0.66): little evidence. With job satisfaction scores $1, 2, 3, 4$ and income scores $7.5, 20, 32.5, 60$ (approximate category midpoints, thousands of dollars), $r = 0.200$ and

$$M^2 = (96-1)(0.200)^2 = 3.81 \quad (P = 0.051),$$
$$M = \sqrt{95}\,(0.200) = 1.95 \quad (\text{one-sided } P = 0.026).$$

Exploiting the trend gives far stronger evidence than X^2 or G^2.

3.4.3 Monotone trend alternatives to independence

A strictly ordinal analysis uses an ordinal measure such as gamma (Section 2.4.4), whose sample version is approximately normal for large samples with SE from the delta method. Then $z = \hat{\gamma}/\text{SE}$ tests independence against monotone association. For Example 2.17, $\hat{\gamma} = 0.221$ with SE $= 0.117$: $z = 1.89$ (one-sided $P = 0.03$) and the 95% interval $0.221 \pm 1.96(0.117) = (-0.01, 0.45)$. The association is at best moderately positive.

3.4.4 Extra power with ordinal tests

X^2 and G^2 have df $= (I-1)(J-1)$ because the alternative has that many more parameters (the nonredundant odds ratios), and they are designed to detect any pattern in them. The ordinal statistics (M^2, $(\hat{\gamma}/\text{SE})^2$) concentrate on one parameter and have df $= 1$. Since the chi-squared mean equals its df, a value with df $= 1$ lies much farther in the tail than a comparable value with df $= (I-1)(J-1)$, giving a smaller P-value; the power advantage grows with I and J (theory in Section 6.4).

3.4.5 Choice of scores

Cochran: any scores fixed before seeing the data give a valid test, but poor scores give an insensitive one. Linear transformations of scores (e.g. $1, 2, 3, 4$ vs. $0, 2, 4, 6$) leave $|r|$ and M^2 unchanged. Results can depend on the scores when the data are very unbalanced.

Example 3.19 (Maternal drinking and congenital malformations)

	Alcohol consumption (average drinks per day)				
Malformation	0	< 1	1–2	3–5	≥ 6
Absent	17,066	14,464	788	126	37
Present	48	38	5	1	1

With two rows, any distinct row scores give the same M^2.
(a) Column scores 0, 0.5, 1.5, 4.0, 7.0: $M^2 = 6.57$, $P = 0.010$.
(b) Equally spaced $1, 2, 3, 4, 5$: $M^2 = 1.83$, $P = 0.18$.
(c) *Midranks* (average rank of a category's subjects in the full ranking: $(1{+}17{,}114)/2 = 8557.5$, then 24,365.5, 32,013, 32,473, 32,555.5): $M^2 = 0.35$, $P = 0.55$.
Midranks treat the sparse last three categories as nearly equal, putting "1–2 drinks" closer to "≥ 6" than to "0", which is inappropriate.

Prefer scores reflecting distances between categories, and do a sensitivity analysis with two or three sensible choices; equal spacing is a reasonable compromise when labels suggest nothing better. With midrank scores for both variables, r is *Spearman's rho*.

3.4.6 Trend tests for $I \times 2$ and $2 \times J$ tables

For a $2 \times J$ table with row scores $u_1 = 0$, $u_2 = 1$, the covariation $\sum\sum u_i v_j n_{ij} = \sum_j v_j n_{2j}$ is the total of the Y scores in row 2, so M^2 detects a difference between the two row means of the Y scores. With midrank scores this is the large-sample *Wilcoxon–Mann–Whitney* test (with ties): its z^2 equals M^2, and it is asymptotically equivalent to statistics based on concordant and discordant pairs. For an $I \times 2$ table M^2 tests a linear trend in the probability of a response category across rows: the *Cochran–Armitage trend test* (Section 5.3.5).

3.4.7 Nominal–ordinal tables

When one variable is nominal with more than two categories, correlation and gamma are inappropriate; statistics summarizing the variation among the mean scores of the ordinal variable across the nominal categories are used instead (Section 7.5.3).

3.5 Small-sample tests of independence

Motivation. For small n the chi-squared approximations fail. Conditioning on the margins eliminates the nuisance parameters and gives an exact null distribution, the hypergeometric, which underlies Fisher's exact test.

3.5.1 Fisher's exact test for 2×2 tables

Theorem 3.20 (Hypergeometric null distribution)

Under independence, for Poisson, multinomial, or independent binomial sampling, conditional on both sets of marginal totals,

$$p(t) = P(n_{11} = t) = \frac{\binom{n_{1+}}{t}\binom{n_{2+}}{n_{+1}-t}}{\binom{n}{n_{+1}}}, \qquad m_- \leq t \leq m_+, \tag{3.8}$$

where $m_- = \max(0,\, n_{1+} + n_{+1} - n)$ and $m_+ = \min(n_{1+},\, n_{+1})$.

Proof.

Step 1. By Theorem 2.3, conditioning a Poisson or multinomial table on its row totals gives independent binomials $n_{11} \sim \text{bin}(n_{1+}, \pi_1)$, $n_{21} \sim \text{bin}(n_{2+}, \pi_2)$; under independence $\pi_1 = \pi_2 = \pi$.

Step 2. Given the row totals, the table is determined by (n_{11}, n_{21}), and $n_{+1} = n_{11} + n_{21} \sim \text{bin}(n, \pi)$. So

$$P(n_{11} = t \mid n_{+1}) = \frac{\binom{n_{1+}}{t}\pi^t(1-\pi)^{n_{1+}-t}\binom{n_{2+}}{n_{+1}-t}\pi^{n_{+1}-t}(1-\pi)^{n_{2+}-n_{+1}+t}}{\binom{n}{n_{+1}}\pi^{n_{+1}}(1-\pi)^{n-n_{+1}}},$$

and the powers of π and $1-\pi$ cancel, leaving (3.8).

Step 3. The range: $0 \le t \le n_{1+}$, $0 \le n_{+1} - t \le n_{2+}$ give $m_- \le t \le m_+$. ■

Given the margins, n_{11} determines the table. For $H_a\colon \theta > 1$, the P-value is $P(n_{11} \ge t_o)$ for observed t_o (Fisher, Irwin, Yates, 1934–35).

Proposition 3.21 (Orderings of 2×2 tables with fixed margins)

With all margins fixed, the sample odds ratio and the difference of proportions $n_{11}/n_{1+} - n_{21}/n_{2+}$ are increasing in n_{11}, and

$$X^2 = \frac{n^3\big(n_{11} - E(n_{11})\big)^2}{n_{1+}n_{2+}n_{+1}n_{+2}}, \qquad E(n_{11}) = \frac{n_{1+}n_{+1}}{n}.$$

Hence one-sided P-values agree for all three orderings, and $P(X^2 \ge X_o^2) = P(|n_{11} - E(n_{11})| \ge |t_o - E(n_{11})|)$.

Proof. [Supplement]

Step 1. With $n_{12} = n_{1+} - n_{11}$, $n_{21} = n_{+1} - n_{11}$, $n_{22} = n - n_{1+} - n_{+1} + n_{11}$, the numerator of $\hat\theta$ increases and its denominator decreases in n_{11}; the difference $n_{11}/n_{1+} - (n_{+1} - n_{11})/n_{2+}$ is increasing too.

Step 2. $n_{11}n_{22} - n_{12}n_{21} = n_{11}(n - n_{1+} - n_{+1} + n_{11}) - (n_{1+} - n_{11})(n_{+1} - n_{11}) = n\,n_{11} - n_{1+}n_{+1} = n[n_{11} - E(n_{11})]$.

Step 3. The familiar 2×2 form $X^2 = n(n_{11}n_{22} - n_{12}n_{21})^2/(n_{1+}n_{2+}n_{+1}n_{+2})$ follows from (3.5): all four $n_{ij} - \hat\mu_{ij}$ equal $\pm(n_{11}n_{22} - n_{12}n_{21})/n$, and $\sum_{i,j} 1/\hat\mu_{ij} = n^3/(n_{1+}n_{2+}n_{+1}n_{+2})$. Substituting Step 2 gives the display. The mean $E(n_{11}) = n_{1+}n_{+1}/n$ is the hypergeometric mean. ■

3.5.2 Fisher's tea drinker

Example 3.22 (Fisher's tea-tasting experiment)

Muriel Bristol claimed she could tell whether milk or tea was poured first. She tasted 8 cups, 4 of each kind, knowing this, and had to name the 4 milk-first cups (order randomized). A possible result:

	Guess poured first		
Poured first	Milk	Tea	Total
Milk	3	1	4
Tea	1	3	4
Total	4	4	8

Both margins are fixed by design, so (3.8) is the natural null distribution. For $H_a\colon \theta > 1$,

$$P(n_{11}=3)=\frac{\binom{4}{3}\binom{4}{1}}{\binom{8}{4}}=\frac{16}{70}=0.229, \qquad P(n_{11}=4)=\frac{1}{70}=0.014,$$

so $P = P(n_{11} \geq 3) = 0.243$. Such a small experiment cannot establish her ability (though she reportedly did convince Fisher).

3.5.3 Two-sided P-values for Fisher's exact test

For a two-sided alternative, different criteria give different P-values:

(a) sum $p(t)$ over t with $p(t) \leq p(t_o)$: $P = P[p(n_{11}) \leq p(t_o)]$ (the most popular);

(b) sum over tables at least as far from H_0: $P = P(|n_{11} - E(n_{11})| \geq |t_o - E(n_{11})|)$, which equals $P(X^2 \geq X_o^2)$ by Proposition 3.21;

(c) $2\min[P(n_{11} \geq t_o), P(n_{11} \leq t_o)]$, which can exceed 1;

(d) the smaller tail probability plus the largest attainable probability in the other tail not exceeding it (Blaker).

Ordering by a distance such as X^2 extends to $I \times J$ tables. Two-sided tests dominate practice; regulatory guidance (ICH E9) recommends one-sided tests at half the usual level, consistent with two-sided intervals.

3.5.4 Discreteness and conservatism issues

In the tea experiment $n_{11} \in \{0, \ldots, 4\}$, so one-sided P-values can only be 0.014, 0.243, 0.757, 0.986, 1.0; rejecting when $P \leq 0.05$ has actual size 0.014. The test is *conservative*.

Example 3.23 (Randomized test of exact size 0.05)

Reject when $n_{11} = 4$; when $n_{11} = 3$ reject with probability 0.157 (if an independent $U \sim \text{Uniform}(0, 1)$ is below 0.157); otherwise accept. Then

$$P(\text{reject } H_0) = E[P(\text{reject} \mid n_{11})] = 1.0(0.014) + 0.157(0.229) + 0.0 \cdot P(n_{11} \leq 2) = 0.05.$$

With randomization Fisher's test is uniformly most powerful unbiased (Tocher, 1950), but data-unrelated randomization is unacceptable in practice. Report the P-value, or the *mid-P-value* (Section 1.4.5): for the tea data, mid-$P = \frac{1}{2}P(n_{11} = 3) + P(n_{11} > 3) = 0.129$. It no longer guarantees type I error $\leq$ nominal, but is rarely much above it.

3.5.5 Small-sample unconditional test of independence*

For two independent binomial samples only the row totals are fixed. To test $H_0\colon \pi_1 = \pi_2$ with a statistic T (e.g. X^2) and observed t_o, for a common value π compute $P_\pi(T \geq t_o)$ from the product

of two binomial pmfs; since π is unknown, use

$$P = \sup_{0 \le \pi \le 1} P_\pi(T \ge t_o).$$

This *unconditional* test (Barnard) also has true size no greater than nominal.

Example 3.24 (Unconditional versus Fisher)

Table with rows $(3,0)$ and $(0,3)$, row totals $3,3$: $X^2 = 6.0$, the maximum, attained also by $(0,3)/(3,0)$. Their probabilities are $\pi^3(1-\pi)^3$ each, so $P_\pi(X^2 \ge 6) = 2\pi^3(1-\pi)^3$. Since $\pi(1-\pi) \le 1/4$ with equality at $\pi = 1/2$, the supremum is $2(0.5)^6 = 0.031$. The two-sided Fisher test gives $2/\binom{6}{3} = 0.100$.

3.5.6 Conditional versus unconditional tests*

Fisher objected that samples with very different total success counts than observed are irrelevant to judging significance. Berger and Boos take the supremum only over a $100(1-\gamma)\%$ confidence interval C_γ for π,

$$P = \sup_{\pi \in C_\gamma} P_\pi(T \ge t_o) + \gamma,$$

with small γ (e.g. 0.001), keeping the size guarantee. Arguments for conditioning: it eliminates nuisance parameters simply and generalizes widely; the margins carry little information about association (some loss unless $\theta = 1$). Arguments against: increased discreteness, hence conservatism and lower power; and for nonnull inference (intervals) conditioning works only for the odds ratio. Fixing only one margin leaves many more tables in the reference set, so the unconditional test is less discrete, less conservative and more powerful, but computationally heavy. With truly independent binomial samples the unconditional approach is sensible; in a randomized trial on a convenience sample, under no treatment effect the response totals are fixed regardless of allocation, which justifies the hypergeometric and Fisher's test (for testing, not for intervals). When both margins are naturally fixed (tea tasting), Fisher's test is best. The mid-P-value reduces conservatism in any case.

3.5.7 Derivation of exact conditional distribution*

Theorem 3.25 (Multiple hypergeometric distribution)

For independent multinomial rows (row totals n_{i+} fixed) or a single multinomial sample, under independence and conditional on both sets of margins,

$$P(\{n_{ij}\} \mid \{n_{i+}\}, \{n_{+j}\}) = \frac{(\prod_i n_{i+}!)(\prod_j n_{+j}!)}{n! \prod_i \prod_j n_{ij}!}, \tag{3.9}$$

for all tables with the given margins. For 2×2 tables this is (3.8).

Proof.

Step 1. Under H_0, $\pi_{j|i} = \pi_{+j}$ for all i, so the product multinomial probability is

$$\prod_i \Big[\frac{n_{i+}!}{\prod_j n_{ij}!} \prod_j \pi_{+j}^{n_{ij}} \Big] = \frac{\prod_i n_{i+}!}{\prod_i \prod_j n_{ij}!} \prod_j \pi_{+j}^{n_{+j}}. \tag{3.10}$$

It depends on the nuisance parameters $\{\pi_{+j}\}$ only through $\{n_{+j}\}$, their sufficient statistics.

Step 2. The column totals are multinomial$(n, \{\pi_{+j}\})$: $P(\{n_{+j}\}) = \frac{n!}{\prod_j n_{+j}!} \prod_j \pi_{+j}^{n_{+j}}$.

Step 3. Since $\{n_{ij}\}$ determines $\{n_{+j}\}$, the conditional probability is (3.10) divided by Step 2, which is (3.9); the π_{+j} cancel.

Step 4. For a single multinomial, first condition on the row totals (Theorem 2.3) to reach Step 1. For 2×2, $\frac{n_{1+}!n_{2+}!n_{+1}!n_{+2}!}{n!\,n_{11}!n_{12}!n_{21}!n_{22}!} = \binom{n_{1+}}{n_{11}}\binom{n_{2+}}{n_{21}}/\binom{n}{n_{+1}}$. ■

The result is free of unknown parameters, so exact probabilities can be computed (Fisher's device of conditioning on sufficient statistics).

3.5.8 Exact tests of independence for $I \times J$ tables*

Exact P-values use (3.9): Freeman and Halton sum the probabilities of tables with the observed margins that are no more likely than the observed one; alternatively order tables by X^2 (Yates), or, for ordinal variables, by an ordinal statistic T (correlation, gamma) with $P = P(T \geq t_o)$.

Example 3.26 (Smoking and myocardial infarction in young women)

A case–control study (Shapiro et al.):

	Smoking (cigarettes/day)		
	0	1–24	≥ 25
Control	25	25	12
Myocardial infarction	0	1	3

Given the margins, the only table with stronger positive association has rows $(25, 26, 11)$ and $(0, 0, 4)$. By (3.9),

$$P = \frac{\binom{25}{0}\binom{26}{1}\binom{15}{3} + \binom{25}{0}\binom{26}{0}\binom{15}{4}}{\binom{66}{4}} = \frac{11{,}830 + 1365}{720{,}720} = 0.018.$$

Although there are only 4 cases, there is evidence of a positive association; the exact $P(X^2 \geq 6.96) = 0.052$, ignoring order, is weaker.

Network algorithms (Mehta and Patel) compute such tests; for large problems Monte Carlo sampling of tables with the given margins estimates the P-value. As I, J grow the statistics take more values and conservatism diminishes.

3.6 Small-sample confidence intervals for 2×2 tables*

Motivation. The conditioning that produced Fisher's test also gives, for nonnull odds ratios, a distribution free of nuisance parameters; inverting it gives exact intervals for θ, though discreteness makes them conservative.

3.6.1 Small-sample inference for the odds ratio

For multinomial sampling, the cell probabilities can be reparametrized as $(\pi_{1+}, \pi_{+1}, \theta)$, since $\theta = \frac{\pi_{11}(1-\pi_{1+}-\pi_{+1}+\pi_{11})}{(\pi_{+1}-\pi_{11})(\pi_{1+}-\pi_{11})}$ is a function of π_{11} given the margins.

Theorem 3.27 (Noncentral hypergeometric distribution)

For independent binomial rows (or a multinomial table), conditional on both margins,

$$f(t; n_{1+}, n_{+1}, n; \theta) = P(n_{11} = t) = \frac{\binom{n_{1+}}{t}\binom{n-n_{1+}}{n_{+1}-t}\theta^t}{\sum_{u=m_-}^{m_+}\binom{n_{1+}}{u}\binom{n-n_{1+}}{n_{+1}-u}\theta^u}, \qquad m_- \le t \le m_+, \tag{3.11}$$

which depends on the parameters only through θ. For $\theta = 1$ it is (3.8).

Proof. [Supplement]

Step 1. By Theorem 2.3 it suffices to treat independent binomials $n_{11} \sim \text{bin}(n_{1+}, \pi_1)$, $n_{21} \sim \text{bin}(n_{2+}, \pi_2)$. For $n_{11} = t$, $n_{21} = s - t$ (with $s = n_{+1}$),

$$P = \binom{n_{1+}}{t}\binom{n_{2+}}{s-t}(1-\pi_1)^{n_{1+}}\pi_2^s(1-\pi_2)^{n_{2+}-s}\left[\frac{\pi_1(1-\pi_2)}{\pi_2(1-\pi_1)}\right]^t,$$

after collecting $\pi_1^t(1-\pi_1)^{-t}\pi_2^{-t}(1-\pi_2)^t$; the bracket is θ.

Step 2. Conditioning on $n_{+1} = s$ divides by the sum of these probabilities over admissible u; the factors not depending on t cancel, leaving (3.11). ■

Proposition 3.28 (Monotonicity, conditional ML, and exact intervals)

Let $T = n_{11}$ have distribution (3.11).
(a) $\frac{d}{d\log\theta}E_\theta(T) = \text{Var}_\theta(T) > 0$ and $\frac{d}{d\log\theta}P_\theta(T \ge t) = \text{Cov}_\theta(\mathbb{1}\{T \ge t\}, T) \ge 0$.
(b) For $m_- < t_o < m_+$ the conditional ML estimate $\hat\theta_c$ is the unique solution of $t_o = E_{\hat\theta_c}(T)$.
(c) Cornfield's tail-method interval—$\{\theta_0 : P_{\theta_0}(T \ge t_o) \ge \alpha/2 \text{ and } P_{\theta_0}(T \le t_o) \ge \alpha/2\}$—has coverage at least $1-\alpha$ for every θ.

Proof. [Supplement] Write $\lambda = \log\theta$ and $c_u = \binom{n_{1+}}{u}\binom{n-n_{1+}}{n_{+1}-u}$, so $f(t) = c_t e^{\lambda t}/K(\lambda)$, $K(\lambda) = \sum_u c_u e^{\lambda u}$ (a one-parameter exponential family).

Step (a). $K'/K = E_\lambda(T)$ and $(K'/K)' = K''/K - (K'/K)^2 = \text{Var}_\lambda(T)$, positive since T takes at least two values. For any h, $\frac{d}{d\lambda}E_\lambda h(T) = E_\lambda[h(T)T] - E_\lambda h(T)E_\lambda T = \text{Cov}_\lambda(h(T), T)$; with $h = \mathbb{1}\{T \ge t\}$, a nondecreasing function of T, this covariance is ≥ 0 (Chebyshev's association inequality: $\text{Cov}(h(T), T) = \frac{1}{2}E[(h(T) - h(T'))(T - T')] \ge 0$ for an independent copy T').

Step (b). The conditional log likelihood is $\ell(\lambda) = \lambda t_o - \log K(\lambda) + \text{const}$, with $\ell'(\lambda) = t_o - E_\lambda(T)$ and $\ell'' = -\text{Var}_\lambda(T) < 0$. $E_\lambda(T)$ increases continuously from m_- to m_+ as λ goes from $-\infty$ to ∞, so for $m_- < t_o < m_+$ there is exactly one root, the maximizer.

Step (c). Fix the true θ. By (a), $\theta < \theta_L(t)$ iff $P_\theta(T \ge t) < \alpha/2$, and as in Theorem 1.14(b) the set of such t is an upper tail with P_θ-probability $< \alpha/2$; likewise for $\theta > \theta_U(T)$. The non-coverage probability is at most α. ■

$\hat\theta_c$ differs from the unconditional ML estimate $\hat\theta = n_{11}n_{22}/(n_{12}n_{21})$. Because of discreteness the actual confidence coefficient (the infimum of the coverage probabilities) can exceed $1-\alpha$ considerably; inverting a single two-sided test, or nonnull unconditional tests for independent binomials, gives shorter intervals.

3.6.2 Tea tasting example

Example 3.29 (Exact inference for the tea data)

For Example 3.22, $\hat{\theta}_c = 6.4$. Cornfield's tail-method interval is $(0.2,\ 626.2)$, with confidence coefficient guaranteed ≥ 0.95; very wide because of the tiny sample. Inverting two-sided exact conditional score tests gives $(0.3,\ 306.2)$. (Treating the rows as binomial samples—inappropriate for this design—inverting exact unconditional score tests would give $(0.4,\ 234.4)$.)

3.6.3 Impact of discreteness on exact confidence intervals

Exact methods are "exact" in that the conditional distribution is free of nuisance parameters, but they are conservative; large-sample methods have no error-rate guarantees and can be conservative or liberal. For the tea data, the Pearson test has $P = 0.157$ versus 0.486 for the two-sided exact test, and the large-sample 95% interval for θ is $(0.4,\ 220.9)$ versus Cornfield's $(0.2,\ 626.2)$. Inverting two one-sided hypergeometric tests with the mid-P-value gives $(0.31,\ 309)$: a compromise that usually comes close to nominal coverage and agrees with the exact method as discreteness diminishes.

3.6.4 Small-sample inference for difference of proportions

Conditioning works only when nuisance parameters have reduced sufficient statistics, which for binary data requires odds-ratio parametrizations (Section 6.7.9); it cannot give intervals for $\pi_1 - \pi_2$ or π_1/π_2. The unconditional approach can: for $H_0\colon \pi_1 - \pi_2 = \Delta_0$ reparametrize by $\Delta = \pi_1 - \pi_2$ and a nuisance parameter (e.g. $\gamma = \pi_1 + \pi_2$, $\pi_1 = (\gamma + \Delta)/2$, $\pi_2 = (\gamma - \Delta)/2$), compute $P_{\Delta_0,\gamma}(T \geq t_o)$ from the binomial product, take the supremum over γ, and collect the Δ_0 with P-value above α. This can be quite conservative; inverting a single two-sided test (Coe and Tamhane) is better than two one-sided ones.

3.7 Extensions for multiway tables and nontabulated responses

Motivation. The tests of this chapter are special cases of model-based inference; models extend them to conditional independence in multiway tables and to continuous explanatory variables.

Tests of independence in two-way tables extend to tests of conditional independence in three-way tables (Chapter 6); with models, the methods also apply when some explanatory variables are continuous.

3.7.1 Categorical data need not be contingency tables

Data files usually have one line per subject (e.g. subject, gender, race, education coded 1–3, opinion coded 1–2), and software forms contingency tables when needed. The generalized linear models of the next chapter contain all methods of this chapter as inferences about parameters of simple special cases.

3.8 R lab

Motivation. Base R implements every test of this chapter: `chisq.test` (with standardized residuals), `fisher.test` (exact test, conditional MLE and exact interval), and a few lines for Woolf

intervals, G^2 partitions and M^2.

Woolf interval and exact interval for the Swedish aspirin study.

```
sw <- matrix(c(28, 656, 18, 658), 2, byrow = TRUE)
lor <- log(sw[1,1] * sw[2,2] / (sw[1,2] * sw[2,1])); se <- sqrt(sum(1 / sw))
exp(lor + c(-1, 1) * qnorm(0.975) * se)        # Woolf: 0.85 2.85
fisher.test(sw)$conf.int                       # conditional exact interval
```

Test of independence with residuals and a G^2 partition.

```
rel <- matrix(c(178,138,108, 570,648,442, 138,252,252), 3, byrow = TRUE)
ct <- chisq.test(rel); ct$statistic; round(ct$expected, 1)
round(ct$stdres, 1)                            # 4.5 -2.6 -1.9 / ... / -6.8 0.7 6.3
G2 <- function(m) { e <- outer(rowSums(m), colSums(m)) / sum(m)
  2 * sum(ifelse(m > 0, m * log(m / e), 0)) }
sch <- matrix(c(90,12,78, 13,1,6, 19,13,50), 3, byrow = TRUE)
parts <- c(G2(sch[1:2, 1:2]),
           G2(cbind(rowSums(sch[1:2, 1:2]), sch[1:2, 3])),
           G2(rbind(colSums(sch[1:2, 1:2]), sch[3, 1:2])),
           G2(rbind(c(sum(sch[1:2, 1:2]), sum(sch[1:2, 3])),
                    c(sum(sch[3, 1:2]), sch[3, 3]))))
round(parts, 2); c(sum(parts), G2(sch))        # 0.29 1.36 12.95 8.43; 23.04
```

Linear trend statistic M^2 (expand the table to one score pair per subject).

```
M2 <- function(tab, u, v) {
  X <- rep(u, rowSums(tab))
  Y <- unlist(lapply(seq_len(nrow(tab)), function(i) rep(v, tab[i, ])))
  r <- cor(X, Y); c(r = r, M2 = (sum(tab) - 1) * r^2,
                    P = pchisq((sum(tab) - 1) * r^2, 1, lower.tail = FALSE)) }
js <- matrix(c(1,3,10,6, 2,3,10,7, 1,6,14,12, 0,1,9,11), 4, byrow = TRUE)
M2(js, c(7.5, 20, 32.5, 60), 1:4)                  # r = 0.200, M2 = 3.81
alc <- rbind(c(17066, 14464, 788, 126, 37), c(48, 38, 5, 1, 1))
M2(alc, 0:1, c(0, 0.5, 1.5, 4, 7))["M2"]           # 6.57
```

Fisher's exact test and mid-P (tea tasting).

```
tea <- matrix(c(3, 1, 1, 3), 2)
fisher.test(tea, alternative = "greater")          # P = 0.243, cond. MLE 6.41
fisher.test(tea)$p.value                           # two-sided 0.486
0.5 * dhyper(3, 4, 4, 4) + dhyper(4, 4, 4, 4)      # mid-P = 0.129
```

R reports the interval (0.21, 621.9); the book's software gives Cornfield's (0.2, 626.2).